Combinatorics 5

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Problem 1

$$(f\delta)(x,y) = \sum_{x \le z \le y} f(x,z)\delta(z,y) = f(x,y),$$
$$(\delta f)(x,y) = \sum_{x \le z \le y} \delta(x,z)f(z,y) = f(x,y).$$

Problem 2

Let $P = \{a, b\}$ where $a \le b$. Define $f(x,y) = 1$ if $x = y = b$ and 0 otherwise, and $g(x,y) = 1$ if $x = y = a$ and 0 otherwise. The functions f and g correspond to the matrices $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, and $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. So f, g aren't zero, but fg is.

Problem 3

Let $P = \{a, b\}$ where $a \le b$ and define $f \in I(P)$ by $f(a,a) = 1$, $f(b,b) = 0$, $f(a,b) = 1$, and $f(x,y) = 0$ for all other x, y with $x \le y$. Then f corresponds to the matrix $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. We have that $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, so f is idempotent.

Problem 4

Part (a)

$$\sum_{x \le y}\sum_{k \ge 0}(\zeta - \delta)^k(x,y) = \sum_{x \le y}(2\delta - \zeta)^{-1}(x,y).$$

Part (b)

$$\sum_{x \le y}(\zeta - \delta)^2(x,y).$$

Problem 7

Call the minimum and maximum elements a and b respectively. For each $y \in P$,

$$\sum_{x \leq y} \mu(x, y) = \sum_{a \leq z \leq y} \zeta(a, z)\mu(z, y) = (\zeta\mu)(a, y) = \delta(a, y).$$

Summing over y gives

$$\sum_{y \in P} \sum_{x \leq y} \mu(x, y) = \sum_{y \in P} \delta(a, y) = 1.$$

Problem 8

Define the poset Q as $(P, \leq_1)$ where $x \leq_1 y$ iff $y \leq x$. Then

$$g(y) = \sum_{x \geq y} f(x) = \sum_{x \leq_1 y} f(x).$$

Applying theorem 4.1, we have

$$f(y) = \sum_{x \leq_1 y} g(x)\mu_Q(x, y).$$

Since the interval $[x, y]$ in P is the same set as the interval $[y, x]$ in Q, it holds that

$$\mu_Q(x, y) = \mu_P(x, y).$$

So

$$f(y) = \sum_{x \leq_1 y} g(x)\mu_Q(x, y) = \sum_{x \geq y} g(x)\mu(x, y).$$